Constructing Corpus of Online Political Discourse

ENG 595 Final Paper

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Self-Identification Statements in Online Political Discourse

News aggregation websites such as Reddit provide a treasure trove of material to analyze as part of a "web as corpus" study. This paper will describe features of the popular website and explore a corpus composed of the comments in a collection of politically oriented subreddits. A brief explanation of the website's functionality will help frame the captured linguistic content, and a discussion of the programming and corpus generation for this project will follow. General features of the corpus, referred to as the Political Subreddit Discourse Corpus or PSDC, are provided, as well as a link to the text files and script used in their generation. Finally, register analyses are performed on each subreddit and serve to highlight features of each community.

Introduction

Reddit Structure. Reddit is a public site that allows semi-anonymous users to post links to various forms of content found on the web. Communities called subreddits are formed based on user defined topical focus. Most subreddits are open to any user to browse, post and comment on; communities are responsible for defining what constitutes an applicable post, and what behaviors are acceptable on the subreddit. These rules are enforced by a combination of automated moderators and volunteer users that apply for a moderator position. A subreddit is denoted by the prefix r/ to help distinguish subreddit names from usernames, which are denoted by the prefix u/. Popular reddit posts that meet certain criteria are promoted to "the frontpage" of r/all, a subreddit that every user sees when they first open reddit, meaning that their visibility to the general public is greatly increased. While the criteria are not publicly available, it widely believed to be a function of upvoting, and the effects of reaching the broader audience of r/all is discussed in several of the comment threads studied. Appearing on the front page carries potentially positive and negative consequences for posts and subreddits that achieve that level of

popularity. In addition to subreddit community guidelines, there are also sitewide rules that govern acceptable behaviors across all of Reddit.

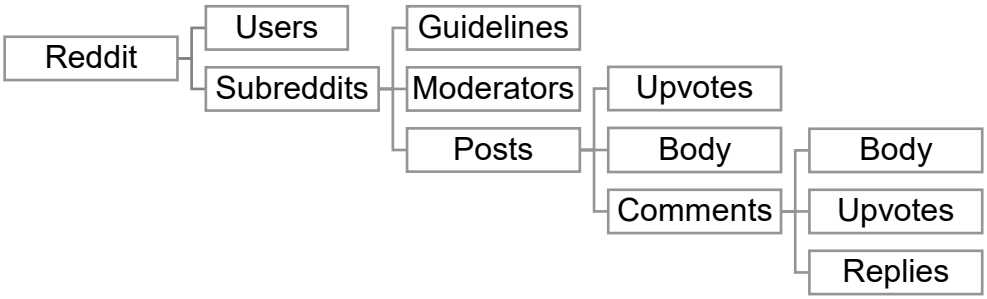


Figure 1. Basics of Reddit Organization. Connections represent a possessive relationship. For example, each subreddit has guidelines, moderators, and posts. Reddit has over 1.2 million subreddits as of January 2018 (redditmetrics.com).

Community Voting. Reddit functions similarly to social media sites like Facebook or Instagram in that central to its model is a user voting system. The Reddit voting system includes an "upvote" (equivalent to a Facebook "like") and a "downvote." These tools allow a user to cast a single vote for every post and comment; however, unlike other social media sites, all voting is anonymous. Votes are used to help filter out content that is irrelevant to the community and popularize content that fits the subreddit's topical focus. This function further distances the reddit voting mechanics from social media sites in that users of a subreddit (ideally) cast their votes with the subreddit guidelines in mind, not based solely on personal opinion. Only moderators have the ability to edit or remove posts and comments, and generally do so in semi-real time.

Methods

Corpus Generation

Comment Compilation Script. Automated access to Reddit is provided to any user that applies through an authentication website; the application process includes creating an identifier

for the script, labelling its purpose, and agreeing to legal terms and conditions. Upon completion, a script is given a unique identifier and access key for use in Reddit's API. An API, or application programming interface, is a collection of functions that allow a third-party application to interact with the supported application. This script uses an API named PRAW, the Python Reddit API Wrapper, an API written for use with the Python programming language. Scripts, also called bots, have access to Reddit in a manner similar to human users, but have additional commands at their disposal and need to follow separate usage guidelines set by Reddit. These guidelines disallow behaviors like vote tampering (automated voting or manipulation of vote counts) and harassment (spamming users with messages, false reports of violations). Unlike subreddit guidelines, script guidelines are sitewide and any deviation from them can result in a ban from the website for that script and any associated accounts.

Script Functionality. To compile the corpus, a Python script utilizing PRAW was written that followed the same process for each subreddit. First, the application logged into reddit through an account registered as a bot. For this paper, the script found the 10 top voted posts of the subreddit and copied up to the first 1,000 comments of each post. The entirety of the script and text files can be downloaded from (note - a current CoyoteMail account is needed): https://drive.google.com/drive/folders/1Ez1_iULHjbLdzY_uHpENs7l0yBysXKup?usp=sharing Each posts' comments were stored in a separate text file and labeled in a manner that allowed for each subreddit to be treated as a sub-corpus. The rationale behind this extraction method is that by using the Reddit voting system, one could take advantage of the community's filtering to find text that is most representative of that community's textual output. Limits were set for the number of posts and comments in order to account for possible differences in the quantity of comments, although the results were mixed in that regard, these results are discussed below.

Political Subreddit Discourse Corpus

Description

Composition. PSDC pulls all of its material from four politically oriented subreddits: r/Conservative, r/Liberal, r/NeutralPolitics and r/politics. As their name suggests, the first two subreddits have an open bias, while the latter two purports to being unbiased. The PSDC's sub-corpora are denoted by the name of the subreddit whose comments populate it. Due to the guidelines of these subreddits, most post bodies are composed solely of links to articles, images or other non-textual content; the media type for each post is shown in Table 1. Two notable observations from this table are that articles dominate a majority of the top posts in most subreddits. Secondly, r/Conservative showed the most variation in post types; it also was the only subreddit that consistently upvoted social media posts like tweets and memes. For summaries of the topic for each post, see the table in Appendix A.

Table 1

Post Medium by Subreddit

Subreddit	Article	Image	Link	Meme	Social Media	Text	Video
r/Conservative	1	2	0	4	2	1	0
r/Liberal	9	0	0	0	0	0	1
r/NeutralPolitics	4	0	2	0	0	4	0
r/Politics	9	0	0	0	0	0	1
Totals	23	2	2	4	2	5	2

Balance of Sub-Corpora. Table 2 summarizes the size of each subcorpora and provides count for the number of unique words (Types) and total number of words (Tokens). The type total is less than the sum of each subreddit's count, since many the types are not unique to a single subreddit. One feature of the frequency measures to take note of is the lack of balance between the number of comments in each subcorpora. This variation in comment and word count

is at least partially due to the difference between (approximate) subreddit populations: r/Conservative has 219,000 subscribers, r/Liberal has 68,400 subscribers, r/NeutralPolitics has 273,000 subscribers and r/politics has 5.2 million subscribers. As detailed in the community voting section, posts that receive a larger number of upvotes are more likely to reach a larger audience through appearing on the front page of r/all. A community with more subscribers has a larger pool of potential upvotes than one with less and as a result, it is more likely receive the requisite number of upvotes to reach r/all. Once a post reaches r/all, it is more likely to be engaged with by the general reddit community, further inflating a post's number of comments.

Table 2
Word Counts for Sub-Corpora

	Token Count	*Type Count*	*Number of Comments*	*Average Tokens/Comment*	*Average Types/Comment*
r/Conservative	166,169	13,519	3,845	43.22	3.52
r/Liberal	33,066	5,517	1,097	30.14	5.03
r/NeutralPolitics	230,622	14,183	3,727	61.88	3.81
r/politics	268,798	18,979	8,847	30.38	2.15
Totals	698,655	32,456	17,516	41.41	3.62

Comparative Analysis

Specialized to General Comparison. Due to the political nature of these subreddits, one of the first questions to test was whether there would be a higher frequency of political terms in the PSDC than in a generalized corpus. To determine whether there was a difference, the ten most frequent content nouns in the PSDC were compared to the same words in Mark Davie's CoCA and iWeb Corpus (Davies n.d.). The results are summarized in Table 3. Frequencies are normalized to 500,000 and the highest frequency for each noun is marked with an asterisk. While not all of the included words are overtly political, the relative prominence of "Trump",

"government", "president", "Russia", and "news" point to the likely nature of discourse occurring in the comments as political.

Table 3

10 Most Frequent Nouns in Comments

Noun	*Absolute Frequency in OPD*	*Frequency in OPD*	*Frequency in CoCA*	*Frequency in iWeb Corpus*
trump	4076	2917.03*	51.01	39.45
people	2802	2005.28*	855.85	640.80
time	1065	762.18	796.30	900.77*
other	1062	760.03	759.75	809.02*
right	1025	733.55*	562.53	363.24
government	1020	729.97*	211.13	123.82
president	967	692.04*	308.74	88.94
russia	828	592.57*	33.22	17.39
news	769	550.34*	159.15	91.15
something	728	521.00*	324.83	251.37

Comparisons between Subcorpora. Variation between subcorpora is another area that seemed ripe for investigation, but also more likely to be affected by post topic. Appendix A summarizes the post body's content for each post that was scraped for comments. By summarizing the posts, it was hoped that some clues might be gathered about variation between subcorpora. Unsurprisingly, President Trump dominated the conversation in three of four subreddits, a phenomenon likely driven by his consistent coverage in news media and the high frequency of news articles as post topics.

Table 4

5 Most Frequent Nouns for each Sub-Corpora

r/Conservative		r/Liberal		r/politics		r/NeutralPolitics	
Noun	*Freq.*	Noun	*Freq.*	Noun	*Freq.*	Noun	*Freq.*
people	162.30	people	194.16	Trump	251.90	Trump	128.39
Trump	111.57	Trump	194.16	people	103.91	people	98.60
Hillary	46.04	trans	60.79	president	75.34	comment	56.72
women	38.82	war	58.07	government	47.77	government	53.07
election	38.64	money	44.46	russia	47.77	comey	50.08

People's consistently high frequency may take more time to explain, as its usage varies. Take the following two example comments (emphasis added):

> [1] Hopefully the same **people** who were on a crusade for Hillary Clinton's head for the exact same thing will also be carrying their torches ad pitchfork for this BLATANT act of indiscretion. (comment 889 from post 5 of r/politics)

> [2] This is bad, **people**. Think about what this means and can be done to any of us on Reddit. An administrator can change any comment we make…to make it say anything they want. […] (comment 3 from post 5 of r/conservative)

In [1] the commenter uses *people* to identify a group that the commenter does not belong to, which can be seen in the derisive mob imagery of pitchforks and torches. In contrast to the exclusionary tone of [1], the commenter in [2] addresses other members of the subreddit in an inclusionary manner, since they clearly identify as part of the people that should be concerned. *People* proves to be a diverse placeholder for sets of the population that commenters may or may not identify as belonging to. We see another use case for people when used to indicate an unnamed source for an article:

> [3] Federal prosecutors have issued grand jury subpoenas to associates of former National Security Advisor Michael Flynn seeking business records, as part of the ongoing probe of Russian meddling in last year's election, according to **people** familiar with the matter. (comment 32 from post 10 of r/NeutralPolitics)

In comment [3] the commenter quotes an article as part of a defense of a claim that a recent presidential decision was politically motivated. *People* occurs as an indication of multiple sources, while also allowing a level of anonymity often required by said sources.

Register Analysis

While Reddit contains a great deal of variation between subreddits, the site possesses some general characteristics describable via a register analysis. Table 5's analysis uses the framework prescribed by Biber, as explained by Crawford and Csomay (2016). Table 6 then explores inter-subreddit variations that were observed between the four subreddits studied.

The Reddit Register	
Participants	All participants in a reddit comment chain must have an account. An account's post and comment history are publicly available and easily checked, but personally identifiable information is not shared. Each account has karma, which is determined by the number of upvotes received from other users. Karma functions as form of reputation, and many subreddits have minimum karma requirements before a user can post or comment within that subreddit.
Relationship Among Participants	Power dynamics in a reddit discussion largely revolve around the giving and receiving of upvotes for posts and comments. Any single user does not have large sway over any other user. However, the speaker (user writing a post or comment) must be mindful of the audience (subreddit receiving the post or comment) because a negative reaction from a large portion of the audience can have a dramatic impact on a user's karma. Additionally, some users have moderator powers to police the interactions in a subreddit. However, moderation actions and frequency vary greatly between subreddits and is not guaranteed to be transparent.
Channel	All interactions occur in the written medium, but in some ways the content appears closer to spoken language. There is frequent use of slang, contractions, misspellings (akin to mispronunciation), and dialectal markers (like *finna* or *yeet*). Text is semi-permanent but can be edited or removed by either users or moderators. Posts or comments that have been edited are publicly labelled as such, and it is common practice for the user to leave a note in the edited body about what edits have occurred as a gesture of transparency and communal good faith.
Production Circumstances	As noted above, text may be edited or removed entirely. How it is produced can vary greatly from user to user, and also based on which subreddit is the target audience. Subreddits that seek to provide meaningful discussion often require responses to be backed by reason or proof, requiring more effort to formulate text. Other subreddits have informal lexicons that allow a user to quickly formulate a response or riff off of another comment or the post being responded to.
Setting	Posts and comments are generated by users through interaction with the main website or a mobile app. Posts are active for a year after their conception but may be locked before that time by a moderator. There is no physical space in which these texts are produced, as reddit is a fully digital medium. Users can access reddit from any physical location that allows them access to the internet.
Communicative Purpose	The communicative purpose varies wildly between subreddits and even between individual posts in a subreddit.
Topic	While reddit is organized roughly by topic, subreddits can accept a broad range of topics, for example the subreddit r/AskReddit allows users to ask any kind of question to the entire reddit community, while the subreddit r/CatsStandingUp is incredibly strict, allowing only media depicting its title, and text may only be one word: "cat".

Table 5

Due to its unique blend of mixture social media components, anonymous chat board style accounts, and link aggregation focus, Reddit may arguably define its own register, with each subreddit further defining the register.

Register Variation Between Political Subreddits

Differences between the political subreddits that form the PSDC appear sharpest in three of the situational characteristics: participants, relationship among participants, and communicative purpose. Table 6 explores these differences, where possible directly quoting from the subreddit's community guidelines, which can be found by visiting that subreddit.

Table 6

	Participants	Relationships	Communicative Purpose
r/Conservative	From subreddit description: "This [subreddit] is by conservatives and for conservatives"	"Leftists and moderates have never been welcomed here. If you wander in here and spout nonsense or insult us, don't be surprised when we ban you almost instantly."	"We are here to discuss conservative topics from a distinctly conservative point of view"
r/Liberal	No clear statement directly from subreddit. Most posters identify as liberal.	"Troll, bot, novelty, throwaway, and alternate accounts are subject to an immediate ban without a warning."	Discussion of news articles from various sources, Q/A style posts.
r/NeutralPolitics	Actively encourages comments from both ends of the political spectrum.	Differences are accepted as long as claims are sourced and arguments do not include personal attacks. Banning not mentioned in community guidelines, mods main action is removal of errant posts and comments.	A heavily moderated community dedicated to evenhanded, empirical discussion of political issues. Based on facts and respectful discussion.

r/politics	No stated political leaning, any participant that can follow rules for civility and post guidelines welcomed.	Personal attacks are always against our rules. Name calling, ad-hominem, demeaning, inflammatory, or other uncivil comments directed at other users are not allowed. Users who break this rule may have their comment(s) removed and be warned and/or banned.	"serious political discussion forum […] for current and explicitly political U.S. news."

Discussion

There are several avenues for improvement of the PSDC and further investigations into corpus features that would likely prove fruitful.

Corpus Construction

The PSDC contains only four of many political subreddits and focuses entirely on US politics. It also favors content from election years, as those times tend to be the subreddits most active; resulting in higher upvote counts for posts. A more robust corpus would need to incorporate both a wider range of the available political subreddits and ideally a wider time range of posts analyzed. Issues of balance must be addressed in such a way as to not completely strip the corpora of it size but have comment counts within a single magnitude of ten. Additionally, the use of Laurence Antony's ant series of software has revealed several gaps in the python scripts tagging methods, including head of file length, date and URL tags that would become necessary to handle larger datasets. Analyses of the PSDC is also hampered by the lack of POS tags, and implementation of tagging software would be necessary to adequately address POS inquiries. Ultimately, the PSDC may serve as a proof of concept for the potential construction of a larger dataset with a broader base of input.

Further Investigations

Several hypotheses could be tested in the PSDC's current state. Linguistic structures that function as self-identification tags in comments represent a phenomenon that seemingly occurs with a greater frequency in Reddit discourse than elsewhere on the web. Political buzzwords like impeachment, alt-right, fake news, and Russian interference presumably occur with great frequency in online discourse, but do not appear in the top five words of any subreddit. Discovering whether this is a factor of corpus construction or lack of instances would possibly provide invaluable feedback for refining the corpus. Cross comparison examinations of the same topic, or even article, may uncover further differences between the conservative and liberal subreddits. By implementing sentiment analyses tools, the corpus could potentially reveal that supposedly neutral subs have sub-conscious ideological bents to them (a belief strongly adhered to by many in r/conservative).

Appendix A

Summary of Post Topics

Subreddit	Post 1	Post 2	Post 3	Post 4	Post 5
r/Conservative	[meme] DNC is hypocritical for criticizing Russian election manipulation	[FB post] Liberal's ad hominem attacks on conservatives drove 2016 electoral loss	[tweet] Rebuttal of a tweet critical of USA; first amendment touted as reason criticism is invalid	[meme] Report of trans-woman victory in athletic event overlaid with image claiming the athlete is still a man	[report] Screenshot of reddit CEO apologizing for editing comments that attacked him
r/Liberal	[article] More GOP politicians have been arrested for misconduct in bathrooms than trans people	[op-ed] Trump's hawkish comments and war nostalgia are dangerous and ill-informed	[article] During interview, Jared Kushner potentially admitted that Trump believes his base is unintelligent	[article] VP Pence's walkout over kneeling in NFL games was staged and fiscally irresponsible	[article] Adult Actress to donate lawsuit winnings to Planned Parenthood in Trump's name
r/NeutralPolitics	[text] Meta discussion about removing the subreddit from the site's main page due to increase in bad behavior	[text] A discussion thread about Net Neutrality after FCC chairman announced potential changes to FCC policy to loosen restrictions on ISPs	[text] A discussion thread focused on fact checking statements from the first presidential debate of 2016	[link] Special Counsel Robert Mueller's report. NOTE: no comments were allowed on this post.	[article] Discussion about a WH press briefing in which certain press outlets were excluded
r/politics	[article] Report that clerk that refused to sign same-sex marriage licenses was voted out of her position.	[article] Report that Trump revealed classified information to Russian foreign minister and ambassador	[article] Report that Trump attempted to have WH counsel McGahn fire special counsel Robert Mueller, McGahn threatened to quit.	[video] Video of Jon Stewart's testimony to Congress on behalf of 9/11 first responders seeking healthcare.	[article] Report that Ivanka Trump used a personal email to transmit government communications in her official capacity

Summary of Post Topics cont.

Subreddit	Post 6	Post 7	Post 8	Post 9	Post 10
r/Conservative	[meme] Both 2016 candidates were undesirable, and their appeal stemmed from dislike of opponent	[image] Homage to Kendrick Castillo, a school shooting victim that attempted to subdue gunman	[article] Report of artist that was kicked from Instagram for painting a lewd mural of Hillary Clinton	[meme] Democrats know they lost the 2016 election from abandoning the Midwest, but still blame Russian interference.	[text] Age restrictions exist against drinking alcohol, having sex, buying mature games, and working, but not gender transition.
r/Liberal	[article] Report of Doug Jones defeat of incumbent Roy Moore for Alabama Senate Seat	[video] Trump displays potentially anti-Semitic sentiment by implicitly accepting belief that recent threats were fake.	[article] Retired General criticizes Trump for skipping veteran's event during trip to France	[article] Polls show Trump has historically low approval rating after a year in office.	[article] Exposé by New York Times that showed Trump significantly downplayed financial assistance from his father
r/NeutralPolitics	[article] Discussion of potential perjury by ex-AG Jeff Session's testimony during senate confirmation	[link] Discussion of linked proposal by House Republicans to replace the Affordable Care Act	[article] Discussion of Donald Trump Jr.'s emails about a Russian proposal to provide incriminating evidence against Hillary Clinton	[text] Reminder that the 2018 midterm elections were following day. Collection of resources to inform voters in the next day's election.	[article] Discussion of Trump's firing of James Comey and analysis of presidential statements.
r/politics	[article] Report that a petition for FCC chairman Pai to step down is increasing in number of signers.	[article] Report that the FBI had opened an inquiry to investigate whether Trump worked for Russia in a secret capacity	[article] Report that FCC's announcement on net neutrality failed to mentioned 22 million comments filed against the pending decision.	[article] Report that Sen. McConnel blocked a vote to fund the federal government for a second time	[articles] Aggregation of articles discussing revelation that Trump directed personal attorney to lie during testimony

Appendix A Notes

Brackets indicate what format the content is submitted in. Posts are ordered by vote counts, meaning Post 1 is the post with the highest upvote count in that subreddit. For non-text or article posts, an attempt is made to distill the message of the image or meme. Posts marked [article] link to news sites including the New York Times and Washington Post, but no distinction was made between news articles and op-eds. Posts marked [text] may contain links but were mainly composes of text written by the poster.

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